

ECP

Event Control Panel

Operator Manual

MagmaLabs · github.com/MagmaSpeedCubes/event-control-panel

Contents

Announce

Audio

Control Panel

Visuals

Settings

Announce

The Announce page is the primary screen for making live or recorded announcements during an event. It combines intercom controls, a shared clock widget, a typed on-screen announcement tool, and the full soundboard — all accessible without leaving the page.

Layout

Column	Contents
Left	Intercom panel · Clock/Stopwatch/Timer · Typed Announcement
Right	Soundboard

Intercom

The intercom routes audio from a microphone to a speaker or audio output device.

Modes

Mode	Behaviour
Live	Microphone audio is passed through to the selected output device in real time.
Recorded	Audio is captured while the button is held active. When stopped, the recording is played back once through the output device.

Controls

Control	Description
Vol	Output volume of the intercom audio.
Input	Select the microphone source. Requires browser microphone permission (granted on first use).
Output	Select the speaker or audio interface to route intercom audio to.
Pause music	Pauses the music queue automatically when the intercom is active, and resumes it when stopped.
Fade music	Instead of hard-pausing, fades music out when the intercom starts and fades back in when it stops.
Soundboard cue	Choose a loaded soundboard sound to play automatically — at the start of a live announcement, or immediately before a recorded playback begins. Set to <i>None</i> to disable.

Control	Description
Start/Stop Announcement	Activates or deactivates the intercom. Label changes to Start/Stop Recording when in Recorded mode.

Tip: The Announce and Control Panel pages share all intercom settings. Changes made on either page are reflected immediately on the other.

Clock / Stopwatch / Timer

A shared widget displaying one of three modes. The state is always in sync with the Control Panel clock widget.

Switching modes

Click the **Clock**, **Stopwatch**, or **Timer** tab to switch.

Clock

Displays the current local time, updated every second.

Stopwatch

Control	Description
Start / Pause	Begins or pauses the stopwatch.
Lap	Records the current elapsed time as a lap entry. Only enabled while running.
Reset	Stops and resets the stopwatch; clears all laps.

Lap times are listed below the display in chronological order.

Timer

Set the countdown duration using the **Min** and **Sec** number inputs (editable while the timer is stopped). The display updates live as you type.

Control	Description
Start / Pause	Begins or pauses the countdown.
Reset	Stops and resets the timer to the configured duration.

When the timer reaches zero, the clock panel pulses red as an alarm until reset.

Typed Announcement

Sends a text overlay to the bottom of the [Display Window](#). The overlay appears on top of whatever is currently being shown.

Fields

Field	Description
Text area	The announcement message. Supports multi-line text.
Text color	Colour of the announcement text. Default: white.
Background	Fill colour of the announcement bar. Default: black.
Opacity	Transparency of the background (0% = fully transparent, 100% = solid).
Linger (s)	Seconds before the announcement auto-clears. Set to 0 to keep it on screen until manually cleared.

Buttons

Button	Description
Show on Screen	Sends the announcement to the Display Window. Opens the window if not already open.
Clear	Removes the announcement from the Display Window immediately.

Note: The Display Window must be open for announcements to appear. If it is not, clicking **Show on Screen** will open it first, which may cause a brief delay.

Soundboard (right column)

The Announce page soundboard is **fully shared** with the Audio page soundboard — any sounds loaded on one page appear on both. Volume changes are synced across pages.

Adding sounds

Click the file picker or drag audio files onto it. Supported formats: MP3, WAV, OGG, M4A, AAC, FLAC, and any other browser-playable audio format.

Controls

Control	Description
Soundboard volume	Master volume for all soundboard playback. Synced with the Audio page slider.
★ (Star)	

Control	Description
	Marks a sound as starred. Starred sounds appear in the dedicated Starred Sounds section on the Control Panel for one-click access during a show.
× (Delete)	Removes the sound from the soundboard permanently (does not delete the file on disk).

Click any sound button to play it immediately. Sounds play on the selected output device at the soundboard volume level.

Audio

The Audio page manages background music playback and the shared soundboard. It supports local audio files, YouTube videos and playlists, and Spotify content.

Layout

Panel	Contents
Left	Music queue and playback controls
Right	Soundboard

Music

Adding music

Local files — Use the file picker to add audio files from your computer. Supported formats: MP3, WAV, OGG, M4A, AAC, FLAC, and any browser-playable audio format. Duration is read automatically after loading.

URLs — Paste a link into the URL input and click **Add** (or press Enter). Supported URL types:

Source	Example URL
YouTube video	https://www.youtube.com/watch?v=...
YouTube playlist	https://www.youtube.com/playlist?list=...
Spotify track	https://open.spotify.com/track/...
Spotify album	https://open.spotify.com/album/...
Spotify playlist	https://open.spotify.com/playlist/...

Spotify note: Spotify items show an amber *"Full playback requires a Spotify login"* warning with a direct link to accounts.spotify.com. Log into Spotify in this browser before playback to get full tracks; free accounts receive 30-second previews via the embed player.

YouTube playlists: When a playlist starts playing, ECP expands it into individual track cards. Real titles are fetched in the background.

Playback controls

Control	Description
Autoplay	Starts playback from the current position (or first non-skipped item).
Pause	Pauses the current track.
Previous	Jumps to the previous non-skipped item. For YouTube playlist items, navigates within the active playlist.
Next	Advances to the next non-skipped item.
Loop mode	Off — stop at end of queue. Loop single — repeat the current track indefinitely. Loop all — loop the entire queue.
Shuffle	Randomises the queue order in place.
Play on finish	When unchecked, playback stops after each track rather than advancing to the next.
Audio crossfade	Fades the current track out and the next track in over ~0.3 s. Not available for YouTube or Spotify sources.
Music volume	Channel volume, multiplied by the master volume on the Control Panel.

Queue items

Each item in the queue shows its title, duration (or source badge), and a row of action controls:

Control	Description
Skip checkbox	Marks the item to be skipped by autoplay and navigation. The item remains in the queue but appears dimmed.
► Queue	Queues this item to play <i>immediately after the currently playing track</i> , overriding sequential advance for one step. Click again to dequeue.
 (Breakpoint)	Marks this item as a breakpoint. Autoplay silently stops before reaching a breakpoint item. Manual navigation (clicking, Next, Previous) shows a confirmation dialog before proceeding. The item is highlighted with a red left border.
↑ / ↓	Moves the item up or down in the queue.
 (Delete)	Removes the item from the queue.

Click the item title row to immediately start playback.

Stream source limitations

Source	Crossfade	Volume control	Next/Prev
Local audio	✓	✓	✓
YouTube	✗	✓	✓ (uses YT playlist nav)
Spotify	✗	✗ (embed limitation)	⚠ advances ECP queue, not Spotify playlist

Controls affected by these limitations show amber or orange styling with a tooltip explaining the restriction.

Soundboard

The soundboard is **shared with the Announce page** — sounds loaded here appear there too, and vice versa.

Adding sounds

Use the file picker to load audio files. Same format support as the music queue.

Controls

Control	Description
Soundboard volume	Master volume for all soundboard clips. Synced with the Announce page slider.
★ (Star button)	Stars the sound. Starred sounds appear in the Starred Sounds section at the bottom of the Music panel on the Control Panel for instant access during a show.
× (Delete button)	Removes the sound from the soundboard.

Click any sound button to play it. Sound routes to the selected output device at soundboard volume × master volume.

Control Panel

The Control Panel is the primary operating screen during a show. It provides a dense, no-scroll overview of all active systems in one place: branding, session management, intercom, clock, music mini-controls, and visuals mini-controls.

Layout

The page is divided into two rows.

Top row (fixed height)

Panel	Contents
Logo & Banners	ECP branding, MagmaLabs link, Ko-fi link
Session / Intercom	Preset management and intercom controls
Clock	Stopwatch / Timer / Clock widget

Bottom row (fills remaining height)

Panel	Contents
Music	Now-playing, playback controls, starred sounds
Visuals	Current media, mirror thumbnail, slide notes
Blank	Master volume, session notes, status

Logo & Banners

Click the **Event Control Panel** banner to reload the app. Links to [MagmaLabs](#) and [Ko-fi](#) open in new tabs.

This panel is always displayed in dark mode regardless of the active theme — its colour never changes.

A small **version tag** appears below the byline, formatted as `vYY.MM.N` where `YY` is the two-digit year, `MM` is the month, and `N` is the sequential commit number within that month (e.g. `v26.6.3` = third deployment in June 2026). The version is fetched automatically from the GitHub API on load and cached for the browser session; if the API is unreachable it shows `vYY.MM.?`.

Session

Presets store the complete app state: music queue, soundboard, media queue, all settings, and notes. They are saved as `.ecp` files (JSON).

Control	Description
Loaded preset	Dropdown of all currently imported presets. Select one to apply it immediately.
Export preset	Saves the active preset (or the current session if none is loaded) to a <code>.ecp</code> file. All audio and image data is embedded as base64 data URIs so the file is self-contained.
Export all	Exports all loaded presets as a single <code>.zip</code> archive.
Import preset(s)	Load one or more <code>.ecp</code> or <code>.json</code> files. Multiple files can be selected at once. The first imported preset is applied immediately; subsequent ones are added to the dropdown for later selection.

File size warning: Exporting a session with many large audio or image files can produce very large `.ecp` files since all media is embedded. For live events, consider loading files fresh each session rather than serialising large assets.

Intercom

A condensed version of the intercom controls from the [Announce page](#). All settings are **bidirectionally synced** — changes here are immediately reflected on the Announce page and vice versa.

See [Announce → Intercom](#) for full documentation of each control.

Clock / Stopwatch / Timer

Same widget as the Announce page, fully synced. Clicking tabs or operating controls on either page updates both simultaneously.

See [Announce → Clock / Stopwatch / Timer](#) for full documentation.

Music (mini panel)

A compact version of the Audio page music controls for use during a show without switching pages.

Control	Description
Now playing	Title of the currently playing track.
Progression	Current position / total duration (0:00 / 3:45).
Autoplay	Start playback.
Next / Previous	Navigate the queue.
Pause	Pause playback.
Vol	Music volume slider. Synced with the Audio page.
Play on finish	When unchecked, stop after each track. Synced with Audio page.
Crossfade	Enable crossfade transitions. Synced with Audio page.

Starred Sounds

If any soundboard sounds are starred (via the ★ button in the Audio or Announce pages), they appear here as compact play buttons for quick one-click access without switching pages.

Visuals (mini panel)

A compact view of the current visual media for use during a show.

Control	Description
Current media	Name of the currently displayed item.
Mirror thumbnail	A live miniature preview of what the Display Window is showing. Displays the image or video name for non-image types.
Previous / Next	Navigate the media queue.
Freeze	Locks the Display Window to the current content. Any navigation (Next, Previous, clicking items) updates the mirror and queue locally, but nothing is sent to the Display Window until Freeze is toggled off. On release, the most recent item is sent automatically. Button turns amber and relabels Unfreeze while active.
Hide	Blacks out the Display Window with a full-screen overlay. New content continues to load behind the overlay so it is ready the instant Hide is released. Button turns dark and relabels Show while active. If the Display Window is reopened while hidden, the blackout is restored automatically.
Fade	Enable fade transitions. Synced with Visuals page.

Notes

Control	Description
	Shows the per-item notes for the currently selected item. Editable and synced with the Visuals page.

Blank panel (right)

Control	Description
Status	Shows a live summary: music play state and queue sizes. Also displays export/import progress messages.
Master volume	Global volume multiplier applied to music and soundboard. Does not affect intercom or system audio.
Session notes	Free-text notepad for the operator. Saved with <code>.ecp</code> exports.

Visuals

The Visuals page manages everything shown on the audience display: images, videos, PDFs, PowerPoint slides, and web embeds. It includes a full-featured presenter view with a mirror display, slide notes, and preview cards for adjacent items.

Layout

Panel	Contents
Left	Media queue, playback controls, file/URL input
Right	Presenter view — mirror display, preview cards, presenter notes

Display Window

Click **Open Display Window** to open a separate browser window (`media.html`) that shows the currently selected media full-screen. Position this window on a projector or secondary display.

- The Display Window remembers its position between sessions.
- If closed, ECP reopens it automatically the next time you show a media item.
- The Display Window operates independently — it does not require ECP to be visible.

Freeze and Hide

Two toggle buttons sit alongside **Open Display Window** (and are also available in the Visuals mini-panel on the Control Panel):

Button	Active state	Behaviour
Freeze	Amber — labelled Unfreeze	Locks the Display Window to whatever is currently shown. Subsequent navigation updates the local mirror and queue, but nothing is sent to the Display Window. On release, the most recently selected item is sent immediately.
Hide	Near-black — labelled Show	Covers the Display Window with a solid black overlay. New content continues loading behind it. Releasing Hide reveals the latest loaded content instantly. If the Display Window is closed and reopened while Hide is active, the blackout is automatically restored.

Tip: Use Freeze to hold a slide while preparing the next cue without the audience seeing you navigate. Use Hide for a full blackout between segments.

Adding media

Local files

Use the file picker or drag files onto it. Supported formats:

Type	Formats
Images	JPEG, PNG, GIF, WebP, BMP, SVG, and any browser-renderable image
Video	MP4, WebM, MOV, and any browser-playable video format
PDF	Converted to individual page images on import using PDF.js
PowerPoint	<code>.pptx</code> files are rendered to slide images on import

PDF and PPTX conversion happens entirely in the browser — no server is required. Each page/slide becomes a separate queue item.

URLs

Paste a link into the URL input and click **Add** (or press Enter).

Source	Behaviour	Warning
YouTube video	Embedded via YouTube's iframe player	⚠ Amber — mirror not in sync; autoplay may be blocked
Google Drive file	Embedded via Drive preview iframe	⚠ Amber — file must be publicly shared
Google Slides	Embedded via Slides embed URL	🔴 Red — must be published to the web; slide timing and navigation are controlled by Google, not ECP
Direct image URL (<code>.jpg</code> , <code>.png</code> , <code>.gif</code> , <code>.webp</code> , <code>image item</code> , <code>.tiff</code>)	Loaded as a normal image item	⚠ Amber — displayed from external URL; will break if source goes offline
Remote PDF (<code>.pdf</code>)	Fetches and converted to pages	⚠ Fetch may fail if the server blocks cross-origin requests
Remote PPTX (<code>.pptx</code>)	Fetches and rendered to slides	⚠ Fetch may fail if the server blocks cross-origin requests

Mirror sync warning: YouTube, Google Drive, and Google Slides items render inside iframes. ECP cannot control or synchronise the mirror display for these — the mirror shows the iframe content but the Display Window must be controlled manually. A red warning banner appears on the Presenter mirror when an embed is active.

Playback controls

Control	Description
Autoplay	Starts automatic sequential playback through the queue.
Pause	Stops automatic advance. The current item remains on screen.
Previous	Shows the previous non-skipped item.
Next	Advances to the next non-skipped item.
Loop mode	Off — stop at end of queue. Loop single — repeat current item indefinitely. Loop all — loop the entire queue.
Transition (s)	How long each image or slide is shown before ECP automatically advances. Does not apply to video or embed items (those advance on video end, or not at all for embeds).
Fade transition	Fades the mirror display and Display Window between items.
Mute video audio	Mutes the audio track on video items when sent to the Display Window.

Queue items

Each item shows its name, type badge, and action row. Click the item to display it immediately.

Control	Description
Skip checkbox	Excludes the item from autoplay and navigation. Item remains visible but dimmed.
► Queue	Queues this item to show immediately after the current item finishes, overriding sequential advance for one step. Click again to dequeue.
 (Breakpoint)	Marks this item as a stop point. Autoplay halts before a breakpoint item. Manual navigation (click, Next, Previous, preview cards) shows a confirmation dialog before proceeding. The item is highlighted with a red left border.
↑ / ↓	Reorders the item in the queue.
 (Delete)	Removes the item from the queue.

Source badges

Badge	Colour	Meaning
YouTube	Red	YouTube embed
Google Drive	Blue	Google Drive iframe
Google Slides	Blue	Google Slides embed
Web Image	Blue	External image URL

Per-item notes

Select any item to enable its notes field in the **Selected slide notes** textarea. Notes are free text, saved per item, and included in `.ecp` exports. They are also mirrored to the Control Panel's notes field.

Presenter panel

Mirror Display

Shows a live thumbnail of the currently displayed item: - **Images**: rendered directly. - **Video**: shows the video name (a full video preview is not shown to avoid double-playback). - **Web embeds**: shows the embed iframe. A red banner appears with a warning that the mirror is not in sync with the Display Window.

Preview cards

Two buttons below the mirror show thumbnails of the **Previous** and **Next** items in the queue relative to what is currently shown. Click either card to navigate to that item (breakpoint confirmation applies).

Presenter notes

A separate free-text textarea independent of per-item notes. Use it for speaker cues, run-of-show notes, or any other information the operator needs during the presentation. Not saved in `.ecp` exports.

Settings

The Settings page controls visual appearance and links to project resources. All appearance settings are persisted in `localStorage` and restored on every page load.

GitHub Repository

A banner at the top of the page links to the [Event Control Panel GitHub repository](#). Click it to open the repo in a new tab.

Appearance

Light Mode

Switches between the default dark theme and a light theme.

- **Off (default):** Dark background (`#101010`), light text. Suited for dim environments such as event backstage areas.
- **On:** Light background (`#f0f2f5`), dark text. Suited for well-lit environments.

The logo and branding panel on the Control Panel always renders in dark mode regardless of this setting.

High Contrast

Inverts button colours relative to the background, making interactive controls more visually distinct from panels.

Theme + High Contrast	Button background	Button text
Dark + HC	Light grey (<code>#d8d8dc</code>)	Dark (<code>#111</code>)
Light + HC	Dark charcoal (<code>#1e1e22</code>)	Light (<code>#eee</code>)

Active buttons (green/red for play/pause/announcement states) are exempt — their indicator colours are preserved.

The logo and branding panel is exempt from High Contrast.

Icon Navigation

When enabled, all **action buttons** (play, pause, next, previous, shuffle, etc.) replace their text labels with Font Awesome SVG icons from the [free collection](#).

- Icons respond to all theme changes: light/dark, high contrast, active states.
- The navigation tabs (**Announce**, **Audio**, etc.) are **not** affected — they always show text.

- Utility buttons (up/down/delete in queues, soundboard star/delete, breakpoint, queue-next) always show icons regardless of this setting.

Realistic Buttons

Expands all action buttons into a taller aviation-panel style with LED bar indicators above the label.

State	LED colour
Idle / inactive	Dark green (unlit)
Playing / active / running	Bright green with glow
Paused	Amber with glow

Button body colour is always dark charcoal (`#25272b`) regardless of Light Mode or High Contrast — the LEDs communicate state instead of background colour changes.

Compatible with Icon Navigation mode: the LED bars appear above the icon.

Feedback

A Fillout embed form at the bottom of the page is available for submitting feedback, bug reports, or feature requests directly within ECP.